

ABOUT

I am a Computer Science undergraduate with a strong foundation in software development and artificial intelligence. My interests span deep learning, computer vision, generative models, and data-driven problem solving. I am passionate about applying technology to build impactful solutions and continuously expanding my knowledge across emerging areas in computer science.

EDUCATION

Degree	Institute	Board / University	CGPA/Percentage	Year
B.Tech CSE	Amrita School of Computing	Amrita Vishwa Vidyapeetham, Coimbatore	8.41 (Till 6th Sem)	2022-2026
Senior Secondary	St. Peter's Senior Secondary School	CBSE	86.2%	2022
Higher Secondary	St. Peter's Senior Secondary School	CBSE	89.8%	2020

- EXPERIENCE
- Amazon

SDE Intern — Finance Automation

May 2025 – June 2025

Bangalore, India

– Finance-Document VQA and Field-Value Extraction POC

Tech: Python, AWS, Amazon Bedrock, Textract

* Built a prototype for **Visual Question Answering (VQA)** and **field-value extraction** on tax documents using LLMs/VLMs.

* Boosted accuracy from **80–85%** to **99% overall**, with **97%** success on critical fields.

* Gained hands-on experience in **prompt engineering** (zero-/few-shot, chain-of-thought) and researched on **parameter-efficient fine-tuning with LoRA/QLoRA**, **knowledge distillation** for compressing large teacher models into lightweight students, and **quantization**.

– Data Pipeline Monitoring System

Tech: AWS Lambda, Athena, DynamoDB, S3, EventBridge, CloudWatch, Secrets Manager, SQS/SNS, Docker

* Designed a scheduled monitoring service that queried multiple data sources to check for missing/late records.

* Implemented automated alerting to pipeline owners on discrepancies using **CloudWatch metrics, dashboards, and alarms**.

* Ensured reliability and data integrity across finance automation pipelines.
- RESEARCH EXPERIENCE

• Estimating Soil Moisture from Satellite Data

Tech: PyTorch, TensorFlow, Excel, Pandas, Matplotlib

March 2024 – July 2025

Amrita Vishwa Vidyapeetham, Coimbatore

– Collaborated with INRAE, France under the guidance of Dr. Amit Agarwal, Professor, TIFAC-CORE in Cybersecurity.

– Engaged in research involving remote sensing, agriculture, and machine learning.

– Leveraged advanced satellite data processing and ML techniques to address challenges in soil moisture prediction and plant life cycles.

• Text-Prompted 3D Mesh Character Animation using GNNs and Diffusion Models

Tech: PyTorch, PyTorch Geometric, Trimesh

November 2024 – August 2025

Amrita Vishwa Vidyapeetham, Coimbatore

– Contributed to research on 3D mesh generation using Graph Neural Networks (GNNs) and diffusion models as part of a professional research elective.

– Developed a latent graph diffusion model to overcome limitations in handling meshes with varying topologies.

– Designed a pipeline integrating GNN autoencoders with diffusion models for text-prompted 3D mesh generation.

– Explored applications in animation and game development by creating a versatile, generalizable approach to dynamic mesh generation.

• Lightweight Student Network for NNUNet

Tech: PyTorch, nnU-Net, Knowledge Distillation, Deep Supervision

September 2025 – Present

Amrita Vishwa Vidyapeetham, Coimbatore

– Developing a multi-stage **model compression pipeline** for NNUNet aimed at reducing parameter count, memory footprint, and inference time while maintaining performance.

– Employing a **multi-phase knowledge distillation** strategy: first at the **feature level**, then at the **soft-label level**, guided by deep supervision.

- Designing preprocessing mechanisms to emphasize **contrast-enhanced regions in DCE-MRI scans**, narrowing the search space and simplifying learning.
- Ultimate goal: deliver a compact, efficient NNUNet variant suitable for clinical deployment in low-resource environments without significant accuracy trade-offs.
- **Final Year Project — Self-Driving Cars with Small Language Models** September 2025 – Present
Tech: Qwen-0.5B LLM, Multimodal Encoders (LiDAR + Multi-axis Camera), Edge AI Amrita Vishwa Vidyapeetham, Coimbatore
 - Experimenting with **edge-focused autonomous driving solutions** using a lightweight Qwen-0.5B decoder paired with encoders for multimodal inputs.
 - Designed pipelines for real-time **waypoint prediction, scene understanding, and object detection (bounding box prediction)** from LiDAR and camera data.
 - Exploring two architectures:
 - * **Parallel multi-encoder design** — separate encoders for LiDAR and multi-axis camera, integrated via the decoder.
 - * **Single fusion encoder** — fusing LiDAR and camera into a unified encoder trained on RGB-point cloud data, reducing compute overhead and improving inference speed.
 - Implementing a **safety system** to predict future states of surrounding objects, ensuring generated waypoints are safe to execute — addressing gaps in many current systems.
 - Entire solution optimized for **local, real-time inference** to enable closed-loop autonomous driving on resource-constrained edge hardware.

PROJECTS

- **Deep Fake Detection** Jan 2024 – Apr 2024
Tech: PyTorch, OpenCV GitHub
 - Developed a video deepfake detection system leveraging multiple detectors with unique methodologies.
 - Built an ensemble framework where results were intelligently combined by a meta-model that assigned weights based on each detector's historical performance on a custom dataset.
- **DDPM Image Generation — Deep Learning Coursework** 2024
Tech: PyTorch, Denoising Diffusion Probabilistic Models (DDPMs) Demo Repository
 - Implemented a **DDPM-based generative model** for image synthesis as part of deep learning coursework.
 - Demonstrated the underlying techniques behind **deepfake generation**, enabling fine-tuning on a handful of images of a target person to produce new, realistic samples.
 - Showcased diffusion-based generative modeling and its applications in media synthesis and AI ethics demonstrations.
- **Adobe India Hackathon — Team Starks (Connecting the Dots)** Jan 2025
Hackathon Project / Tech: Qwen2.5-0.5b (Int8, llama.cpp), YOLOv8n GitHub
 - Built an intelligent, lightweight, **CPU-only offline system** to transform static PDFs into dynamic, structured, persona-aware knowledge artifacts.
 - Designed a **layout-aware Small Language Model (SLM)** using **Qwen2.5-0.5b (Int8 quantized)** on **llama.cpp** for efficient low-resource inference.
 - Integrated **2× YOLOv8n models** (distilled with PP-DocLayout-L + a custom outline detector), **SentenceTransformers** for semantic search, and **K-means clustering** for hierarchical structuring (H1, H2, H3).
 - Enabled **semantic retrieval + summarization** by ranking the top 5 relevant sections in embedding space, then summarizing them via the SLM.
 - Optimized the pipeline to meet hackathon size limits (**200 MB in Round 1A, 1 GB in Round 1B**), achieving high portability and efficiency. The system processed 10–15 documents of 50 pages each in under one minute.
- **Fire Fighting Drone for Early Forest Fire Detection and Extinguishment** 2018
School Project / Tech: Drones, Sensors, Fire Suppression Systems
 - Developed a drone capable of early forest fire detection and suppression, with integrated surveillance and rapid response mechanisms.
 - **Achievements:** CBSE Science Fair State Finalist, PPTIA Innovation Award National Finalist (Top 10), First Prize at Shastra Science Fair.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C++, C
- **DeepLearning Frameworks:** Pytorch, Pytorch3D, TensorFlow, Scikit-Learn,
- **Data Analytics Tools:** NumPy, Pandas, Matplotlib, Seaborn
- **Image processing Libraries:** MediaPipe, OpenCV
- **Cloud Platforms:** AWS (Lambda, S3, CloudWatch, Athena, DynamoDB, Bedrock, Sagemaker)
- **LLM Techniques:** Prompt Engineering, Model Fine-tuning, Quantization, Knowledge Distillation
- **Accelerated Computing:** CUDA programming for GPU-accelerated deep learning and high-performance computing

CERTIFICATIONS

- **The Joy Of Computing Using Python || NPTEL IITM || ELITE**

ACHIEVEMENTS

- **1st place in Nestria Jan Built Hackathon** ,Built a deep fake detection website, [Link](#)

April 2024